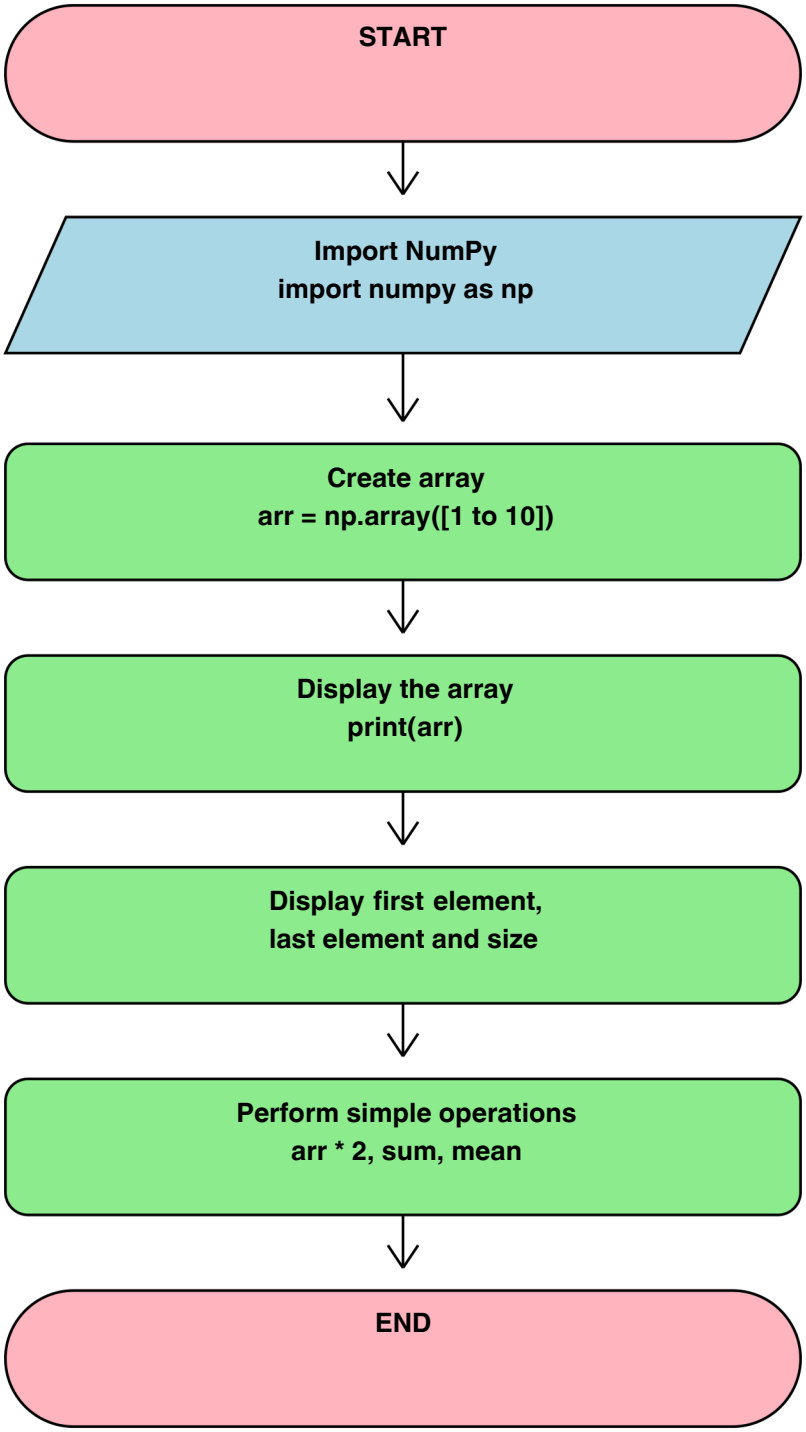
NUMPY ONE-DIMENSIONAL ARRAY

1. Simple Algorithm

1. Start the program.
2. Import the NumPy library using **import numpy as np**.
3. Create a one-dimensional array containing numbers 1 to 10 using **np.array()**.
4. Display the complete array.
5. Display the first element of the array.
6. Display the last element of the array.
7. Display the size (number of elements) of the array.
8. Multiply every element of the array by 2 and display the result.
9. Calculate and display the sum of all elements.
10. Calculate and display the mean of all elements.
11. Stop the program.

2. Flowchart

Flowchartforthesimplified program:



3. Simple Python Program

```
import numpy as np

# Create an array
arr = np.array([1, 2, 3, 4, 5, 6, 7, 8, 9, 10])

# Display the array
print("Array:", arr)

#   Display   basic   information
print("First  element:",  arr[0])
print("Last  element:",  arr[-1])
print("Size of array:", arr.size)

# Simple operations
print("Array multiplied by 2:", arr * 2)
print("Sum of array:", arr.sum())
print("Mean of array:", arr.mean())
```

main.py

Visualize

Run

```python
import numpy as np

# Create an array
arr = np.array([1, 2, 3, 4, 5, 6, 7, 8, 9, 10])

# Display the array
print("Array:", arr)

# Display basic information
print("First element:", arr[0])
print("Last element:", arr[-1])
print("Size of array:", arr.size)

# Simple operations
print("Array multiplied by 2:", arr * 2)
print("Sum of array:", arr.sum())
print("Mean of array:", arr.mean())
```

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Output

```
Array: [ 1  2  3  4  5  6  7  8  9 10]
First element: 1
Last element: 10
Size of array: 10
Array multiplied by 2: [ 2  4  6  8 10 12 14 16 18 20]
Sum of array: 55
Mean of array: 5.5
```

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